

Skin Disease Detection Using Deep Learning

1. Introduction

This project focuses on the development of a deep learning-based system for detecting skin diseases using medical image datasets. The model is trained using the HAM10000 dataset, which contains dermatoscopic images of different skin lesion categories. The main objective of this project is to automate disease detection and improve early diagnosis using artificial intelligence.

2. Objectives

To classify skin diseases using image processing and deep learning techniques. To preprocess and normalize medical images for better model performance. To train a CNN-based model for accurate disease prediction. To evaluate the model using accuracy and prediction metrics.

3. Dataset Description

The HAM10000 dataset was used for this project. It contains thousands of skin lesion images categorized into multiple disease classes. Metadata information such as lesion type and diagnosis was also used during preprocessing and labeling.

4. Technologies Used

Technology	Purpose
Python	Programming Language
TensorFlow/Keras	Deep Learning Framework
NumPy	Numerical Computation
Pandas	Data Handling
OpenCV	Image Processing
Matplotlib	Data Visualization
Scikit-learn	Model Evaluation

5. Methodology

Step 1: Dataset Collection

The HAM10000 dataset was collected and organized for training and testing.

Step 2: Data Preprocessing

Images were resized to 128x128 pixels, normalized, and labeled appropriately. Preprocessed data was stored in compressed NPZ format.

Step 3: Model Building

A Convolutional Neural Network (CNN) model was developed using TensorFlow/Keras for image classification.

Step 4: Training and Testing

The dataset was split into training and testing sets. The model was trained over multiple epochs to improve accuracy.

Step 5: Prediction

The trained model predicts the category of skin disease from a given skin image.

6. Key Features

Automated skin disease classification Medical image preprocessing Deep learning-based prediction system Scalable architecture for future improvements Efficient image handling and classification

7. Results

The model successfully learned to classify different categories of skin diseases from dermoscopic images. The project demonstrates the effectiveness of deep learning in healthcare applications and medical image analysis.

8. Conclusion

This project demonstrates how deep learning can assist in medical diagnosis by automating skin disease detection. The developed system can help healthcare professionals by providing faster preliminary predictions and reducing manual analysis effort. Future improvements may include deploying the model as a web or mobile application and improving accuracy using larger datasets.

9. Future Scope

Deploying the model as a web application Integrating real-time image capture Improving accuracy using transfer learning Adding more disease categories Cloud deployment for remote healthcare access

10. References

HAM10000 Dataset TensorFlow Documentation Keras Documentation Research papers on medical image classification